

Chapter 41

Voice Activity Detection (VAD)

41.1 Introduction

ESP32-P4 integrates a Voice Activity Detection (VAD) module. This module facilitates the hardware implementation of the first-stage algorithm for voice wake-up and other multimedia functions. Additionally, it provides hardware support for low-power voice wake-up solutions.

41.2 Feature List

The VAD module has the following features:

- VAD algorithm processes voice data frame by frame, with each frame containing 256 data points. The data sampling rate is 8 kHz, and the bit width is 16 bits
- 2 KB buffer that stores up to 4 frames of data
- Independent system wake-up source
- Configurable interrupt sources
- Flexible configuration of algorithm parameters

41.3 Architectural Overview

Figure [41.3-1](#) is the block diagram of the ESP32-P4 VAD module. The VAD module includes:

- Energy threshold check
- FFT (Fast Fourier Transform) calculation
- LTSD (Long-Term Spectral Divergence) calculation

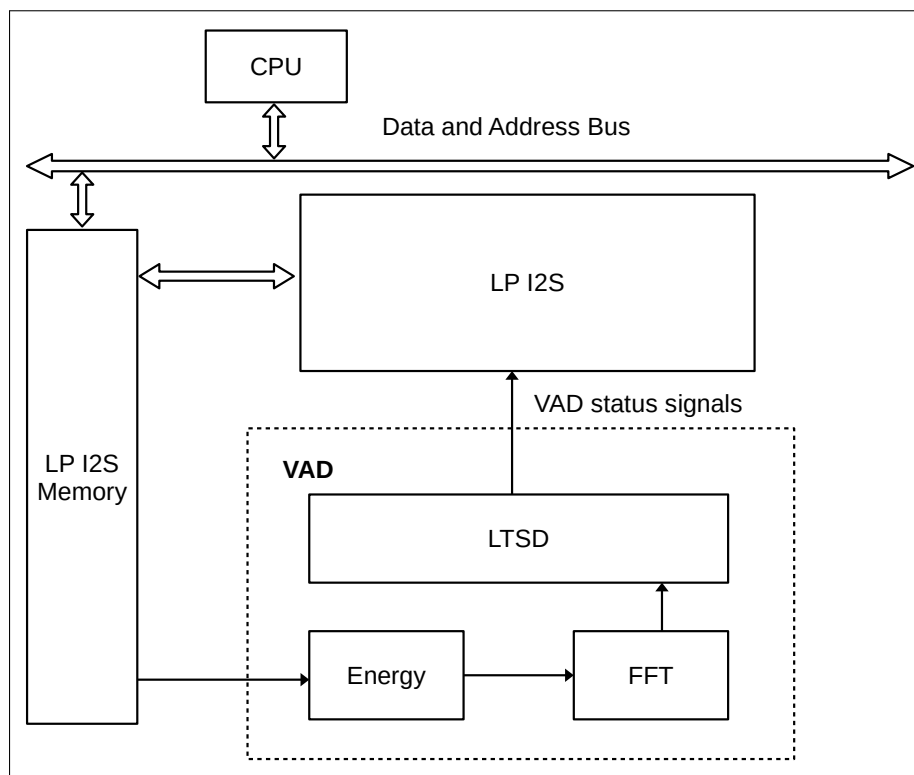


Figure 41.3-1. ESP32-P4 VAD Architecture

The VAD module reads 8 kHz, 16-bit wide audio data from LP I2S memory. It processes the data through three stages: energy detection, FFT, and LTSD. Then the module detects the voice activity status and sends the VAD status signal to the LP I2S module, which is used to generate the corresponding interrupt and wake-up signal.

41.4 Functional Description

The ESP32-P4's VAD module offers flexible and configurable functionality. Users can adjust algorithm parameters and wake-up sources to meet their specific needs.

41.4.1 Algorithm Parameter Configuration

Number of Initial Frames

When the VAD module starts running, it first enters an initialization phase before normal operation. During this phase, it automatically learns noise parameters. The duration of the initialization phase depends on the number of initial frames configured by the user. Proper setting of initial frames can effectively suppress noise and significantly reduce false positives. However, if the initialization phase is extended over too many frames, voice activity within these frames may be missed. Therefore, please configure an appropriate number of initial frames based on actual conditions.

It is recommended to place the device in the target environment noise during the initialization, and set the number of initial frames to 100-200 frames.

Users can configure the number of initial frames via [LP_I2S_PARAM_INIT_FRAME_NUM](#).

Note:

It is recommended to let the VAD module learn noise parameters during the initialization. However, the VAD module will also continuously learn and update noise parameters during normal operation. Therefore, in scenarios where the false positive rate during initial frames is not a concern, the number of initial frames can be reduced, allowing the VAD module to enter the normal operation more quickly.

Energy Threshold

The VAD module sets an energy threshold before the voice activity detection. Setting the threshold can filter out low-energy voice activities (e.g., distant voices) from triggering a wake-up. Appropriate energy threshold setting can effectively reduce false positives and conserve power.

It defines the energy value of a speech frame as the mean square of one-frame data. The VAD module calculates the sum of squares of one-frame data in real-time. If this sum is less than the energy threshold multiplied by 256, subsequent calculations will not be performed. Users can configure the energy threshold via [LP_I2S_PARAM_MIN_ENERGY](#).

Band Energy Check

The band energy check assesses the proportion of band energy within the overall frequency spectrum. Enabling the band energy check in varying environments can help reduce false positives; however, it may also reduce true positives. Users should determine whether to enable or disable the band energy check based on the environment.

Users can enable or disable the band energy check via [LP_I2S_PARAM_SKIP_BAND_ENERGY](#).

- 0: Enable band energy check
- 1: Disable band energy check

Voice Activity Detection

The VAD module has two voice activity status: voice activity status and non-voice activity status, managed by voice frame counter `speech_count` and non-voice frame counter `silent_count`. These counters control transitions between the status as follows:

- **Non-voice activity status:**

- When voice activities are detected in the current frame, the count value of `speech_count` increases by 1.
- When voice activities are not detected in the current frame, the count value of `speech_count` decreases by 1 (down to a minimum of 0).
- If the count value of `speech_count` is greater than [LP_I2S_PARAM_HANGOVER_SPEECH](#), the VAD module enters the voice activity status and resets `silent_count` to 0.
- Under the non-voice activity status, the VAD module continuously learns and updates noise information.

- **Voice activity status:**

- When voice activities are detected in the current frame, the count value of `speech_count` increases by 1, and the count value of `silent_count` decreases by 1 (down to a minimum of 0).

- When voice activities are not detected in the current frame, the count value of `silent_count` increases by 1:
 - * If the count value of `speech_count` is greater than `LP_I2S_PARAM_MIN_SPEECH_COUNT`, and
 - The count value of `silent_count` is greater than `LP_I2S_PARAM_HANGOVER_SILENT`, the VAD module transitions to the non-voice activity status and resets both `speech_count` and `silent_count` to 0;
 - The count value of `silent_count` is less than `LP_I2S_PARAM_HANGOVER_SILENT`, the VAD module maintains the voice activity status.
 - * If the count value of `speech_count` is less than or equal to `LP_I2S_PARAM_MIN_SPEECH_COUNT`, the VAD module transitions directly to the non-voice activity status and resets `speech_count` to 0.
- Under the voice activity status, the VAD module pauses updating noise information. When the count value of `speech_count` exceeds `LP_I2S_PARAM_MAX_SPEECH_COUNT`, indicating stability in the new environment, it resumes learning and updating noise information.

Status Registers

The VAD module provides several key operational variables as status registers (read-only) to facilitate user debugging. The status registers include:

- `LP_I2S_VAD_FLAG`: Represents the current voice activity detection status. 1 represents the VAD module is in the voice activity status, while 0 represents it is in the non-voice activity status.
- `LP_I2S_ENERGY_ENOUGH`: Represents whether the current frame passes the energy threshold check. 1 represents the current frame passes the energy threshold check, while 0 represents not pass.
- `LP_I2S_SPEECH_COUNT_OB`: Represents the count value of the voice frame counter, `speech_count`.
- `LP_I2S_SILENT_COUNT_OB`: Represents the count value of the non-voice frame counter, `silent_count`.

41.4.2 Wake-up Source Configuration

When the VAD module is in the voice activity status, it triggers the wake-up source `LP_I2S_WAKEUP`. Users can configure the wake-up source (please refer to Chapter [47 LP I2S Controller](#)), enabling the VAD to become one of the wake-up sources for ESP32-P4.

41.5 Interrupts

ESP32-P4's VAD module shares the same interrupt signal `LP_I2S_INTR` with the LP I2S module. The following interrupt sources can generate the interrupt signal:

- `LP_I2S_VAD_DONE_INT`: Triggered when the VAD completes processing one frame of data.
- `LP_I2S_VAD_RESET_DONE_INT`: Triggered when the VAD is reset.

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [41.7 Register Summary](#).

41.6 Programming Procedure

41.6.1 Automatic Operation Mode

1. Configure the VAD parameters as described in Section [41.4.1](#).
2. Enable the VAD automatic detection mode by writing 1 to [LP_I2S_VAD_EN](#).
3. Configure the LP I2S to operate in 16-bit receive mode and start receiving data in 16-bit format. The received data will be stored in the LP I2S memory.
4. The VAD module will automatically perform operations on each frame of data after the LP I2S receives the complete frame.
5. (Optional) When the VAD module triggers a wake-up source/interrupt signal, the LP I2S memory already contains the voice frame that triggered the VAD wake-up, along with the previous three voice frames. Users can read these four frames of voice data from the LP I2S memory via the LP core for subsequent processing. For details on the size of the LP I2S memory and data reading methods, please refer to Chapter [47 LP I2S Controller](#) > Section [47.8.3 Internal Memory](#).

41.6.2 Manual Operation Mode

1. Configure the VAD parameters as described in Section [41.4.1](#).
2. Manually store the voice data in the LP I2S memory. For details, refer to Chapter [47 LP I2S Controller](#) > Section [47.8.3 Internal Memory](#).
3. With all the voice data stored in the memory, write 1 to [LP_I2S_VAD_FORCE_START](#), then the VAD will perform operations on one frame of data.
4. When the VAD module completes processing one frame of data, [LP_I2S_VAD_DONE_INT](#) interrupt will be generated.

41.7 Register Summary

The addresses in this section are relative to VAD base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
VAD Registers			
LP_I2S_VAD_CONF_REG	Operation control register	0x0000	varies
LP_I2S_VAD_RESULT_REG	Voice activity status register	0x0004	RO
LP_I2S_VAD_PARAMO_REG	Parameter configuration register 0	0x0080	R/W
LP_I2S_VAD_PARAM1_REG	Parameter configuration register 1	0x0084	R/W
LP_I2S_VAD_OBO_REG	Status register for key operational variables	0x00B0	RO

41.8 Registers

The addresses in this section are relative to VAD base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 41.1. LP_I2S_VAD_CONF_REG (0x0000)

(reserved)																																LP_I2S_VAD_FORCE_START LP_I2S_VAD_RESET LP_I2S_VAD_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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LP_I2S_VAD_EN Configures whether to enable the automatic operation mode.

0: Disable

1: Enable

(R/W)

LP_I2S_VAD_RESET Configures whether to reset the VAD module.

0: Not reset

1: Reset

(WT)

LP_I2S_VAD_FORCE_START Configures whether to trigger the manual operation on one frame of data.

0: Not trigger

1: Trigger

(WT)

Register 41.2. LP_I2S_VAD_RESULT_REG (0x0004)

(reserved)																															LP_I2S_ENERGY_ENOUGH LP_I2S_VAD_FLAG																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
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LP_I2S_VAD_FLAG Represents the voice activity status. For details, see Section 41.4.1. (RO)

LP_I2S_ENERGY_ENOUGH Represents whether the current frame passes the energy threshold check. For details, see Section 41.4.1. (RO)

Register 41.3. LP_I2S_VAD_PARAM0_REG (0x0080)

(reserved)								LP_I2S_PARAM_INIT_FRAME_NUM								LP_I2S_PARAM_MIN_ENERGY								
3125								2416								150								
0000000								200								5000								Reset

LP_I2S_PARAM_MIN_ENERGY Configures the energy threshold. For details, see Section 41.4.1. (R/W)

LP_I2S_PARAM_INIT_FRAME_NUM Configures the number of initial frames. For details, see Section 41.4.1. (R/W)

Register 41.4. LP_I2S_VAD_PARAM1_REG (0x0084)

LP_I2S_PARAM_SKIP_BAND_ENERGY																(reserved)								LP_I2S_PARAM_HANGOVER_SILENT																LP_I2S_PARAM_HANGOVER_SPEECH																LP_I2S_PARAM_MAX_SPEECH_COUNT																LP_I2S_PARAM_MIN_SPEECH_COUNT															
31		30																						24		23		16										15		11						10		4										3		0																											
0		0x0										30										3				100										3				Reset																																															

LP_I2S_PARAM_MIN_SPEECH_COUNT Configures the transition between voice activity status. For details, see Section 41.4.1. (R/W)

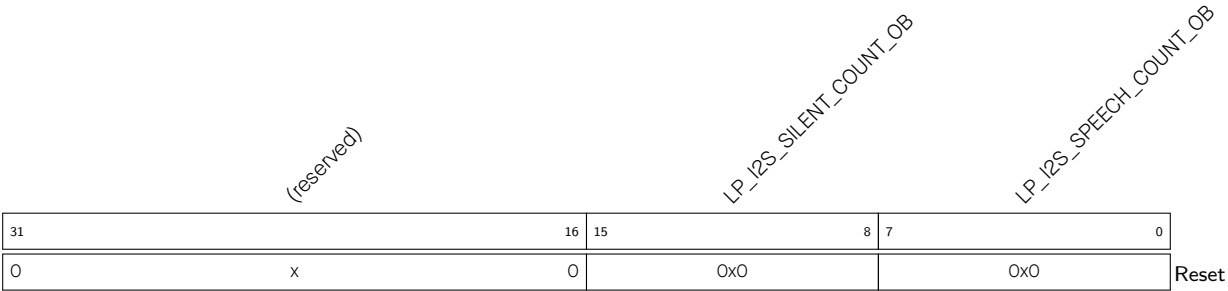
LP_I2S_PARAM_MAX_SPEECH_COUNT Configures the transition between voice activity status. For details, see Section 41.4.1. (R/W)

LP_I2S_PARAM_HANGOVER_SPEECH Configures the transition between voice activity status. For details, see Section 41.4.1. (R/W)

LP_I2S_PARAM_HANGOVER_SILENT Configures the transition between voice activity status. For details, see Section 41.4.1. (R/W)

LP_I2S_PARAM_SKIP_BAND_ENERGY Configures the transition between voice activity status. For details, see Section 41.4.1. (R/W)

Register 41.5. LP_I2S_VAD_OB0_REG (0x00B0)



LP_I2S_SPEECH_COUNT_OB Represents the count value of the voice frame counter, speech_count. For details, see Section 41.4.1. (RO)

LP_I2S_SILENT_COUNT_OB Represents the count value of the non-voice frame counter, silent_count. For details, see Section 41.4.1. (RO)